Einführung in MLOps

05b Infrastructure Exercise Setup

Tobias Mérinat

teaching2025@fsck.ch

Lucerne University of Applied Sciences and Arts
HOCHSCHULE LUZERN

Department of Information Technology
Lucerne University of Applied Sciences and Arts
6343 Rotkreuz, Switzerland

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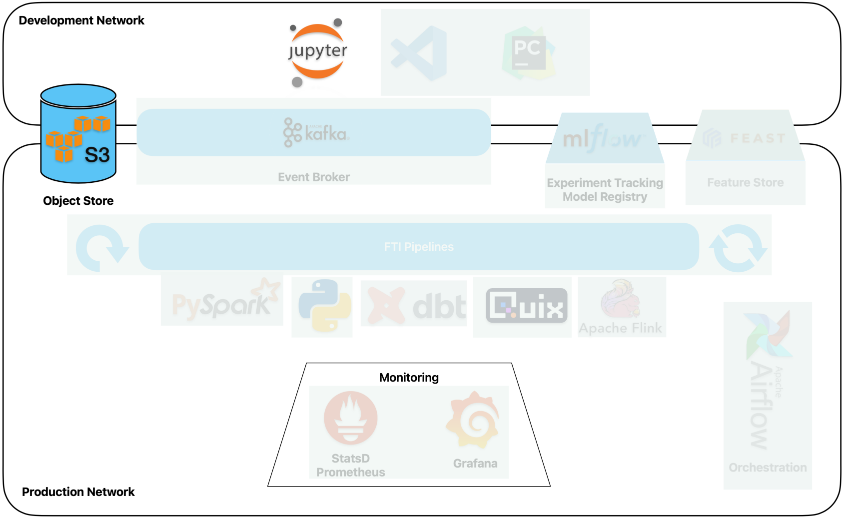
Object Store and Development Environment



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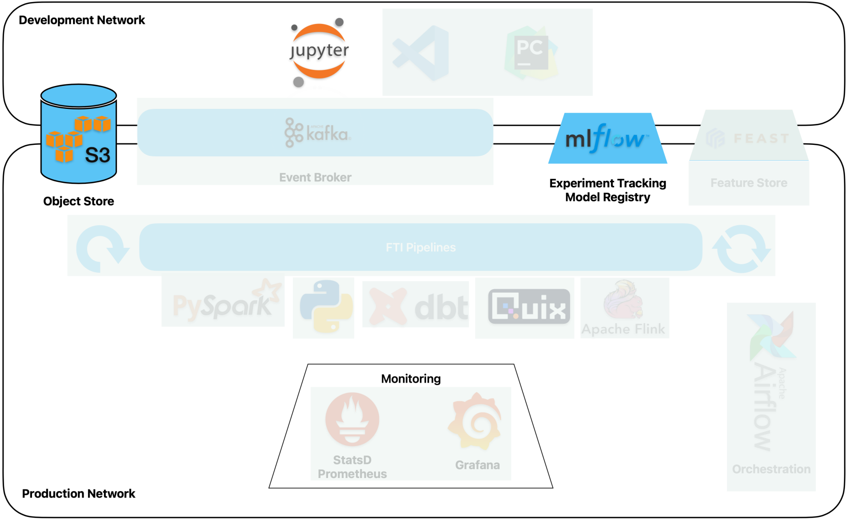
Model Registry (MLFlow)



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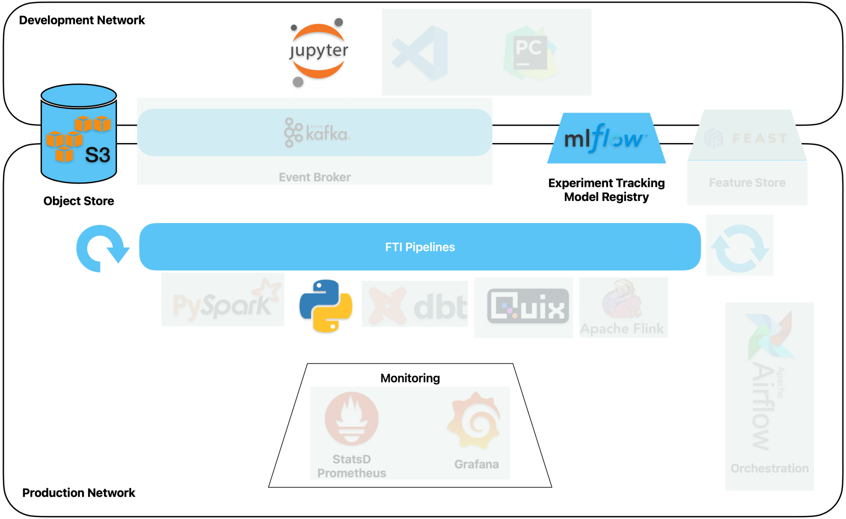
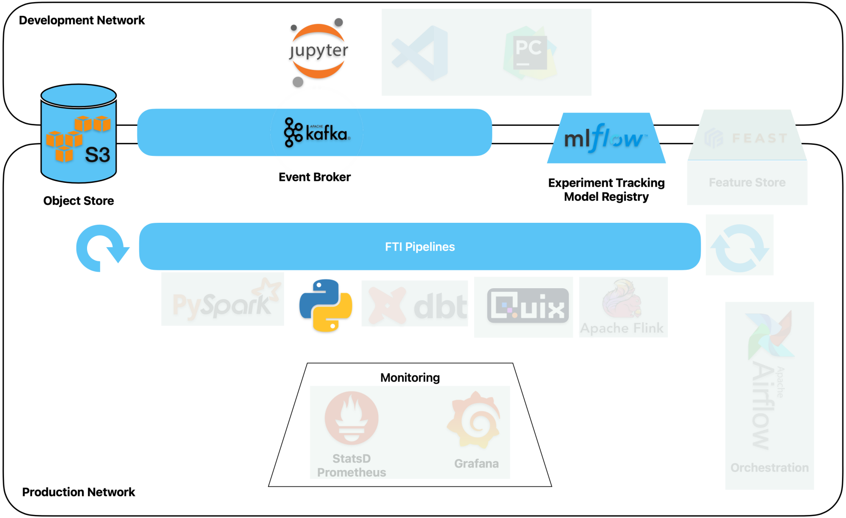
Batch Pipeline with Python



Event Streaming Platform (Redpanda)



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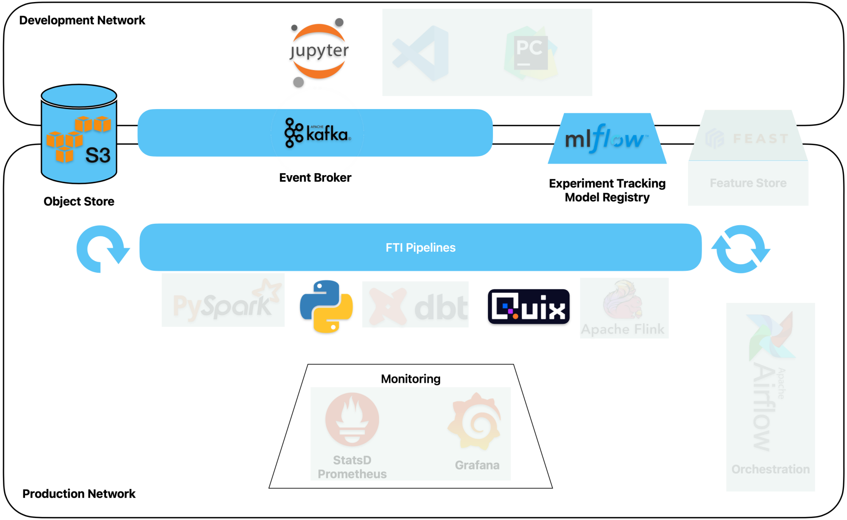
Real time Pipeline with Quix



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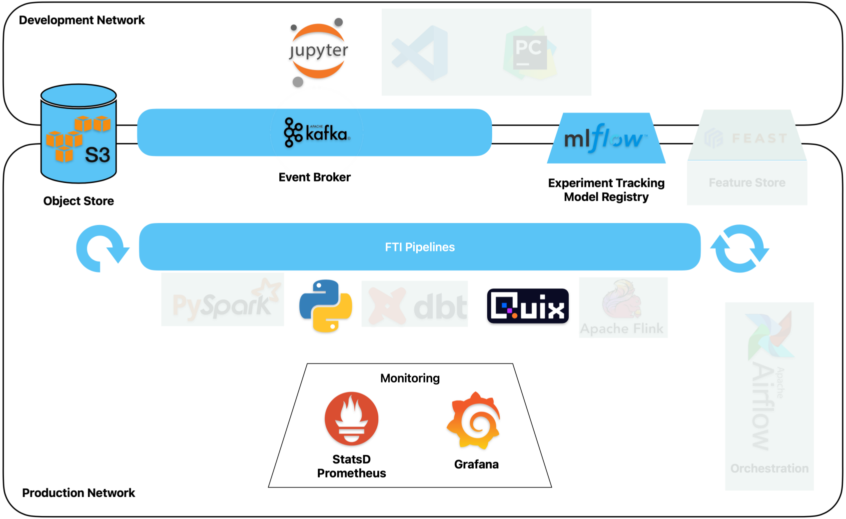
Monitoring (Prometheus / StatsD / Grafana)



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Infrastructure is containerized

- All services run inside docker containers
- When you set up services, configure, start or stop them you are working on the host system (i.e. **not** inside a container)
- You **do not** build models or execute code on the host system, you always do this inside a container
  - In exercises that are about setting up infrastructure, you work outside of containers
  - In exercises where you build models, pipelines, run them, or access web frontends (jupyter, mlflow, grafana, . . .) you work inside containers

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Available containers

- The following two containers are pre-configured:
  - Object Store (*objectstore/docker-compose.yml*)
  - Jupyter Development Environment (*development_env/docker-compose.yml*)
- And these six will be set up by you in the course of the exercises:
  - MLFlow model registry and experiment tracking (*mlflow/docker-compose.yml*)
  - Redpanda (kafka-compatible) message broker and console (*message-broker/docker-compose.yml*)
  - Prometheus, StatsD and Grafana for monitoring (*monitoring/docker-compose.yml*)
- For simplicity reasons, pipelines do not run in separate containers. But they should. . .

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Github Codespaces

Codespaces main window